SOLUTION ARCHITECTURE & RELATIONAL SCHEMA

Detailed Database Entity Schema and System Component Layout

1. System Components

The VBCUA is structured into three main layers:

1. **Presentation Layer**: Managed by `app.py`, which renders the Streamlit pages, collects configuration parameters (silence thresholds, model sizes), displays interactive analysis cards, and handles file download streams.

2. **Business Logic Layer**: Located under `modules/` which isolates transcription, semantic evaluation, audio feature extraction, and report creation.

3. **Database Layer**: Handled by `database.py` which defines SQLAlchemy tables and manages sessions.

2. Relational Schema Tables

• **USER**: Stores user details (username, email).

• **SESSION**: Tracks active sessions linked to users.

• **AUDIO_FILE**: Stores filename, file size, filepath, and duration.

• **REFERENCE_CONCEPT**: Stores reference concepts (Photosynthesis, Neural Networks, etc.).

• **TRANSCRIPT**: Stores transcribed text and word count.

• **FILLER_WORD_STATS**: Stores total filler word counts, ratios, and JSON details.

• **SEMANTIC_SIMILARITY**: Stores the raw Sentence-BERT similarity score.

• **AUDIO_FEATURE**: Stores extracted Librosa metrics (pause ratio, rms energy).

• **EVALUATION_RESULT**: Integrates overall score, category, and feedback notes.

• **REPORT**: Tracks pdf filepaths and creation timestamps.